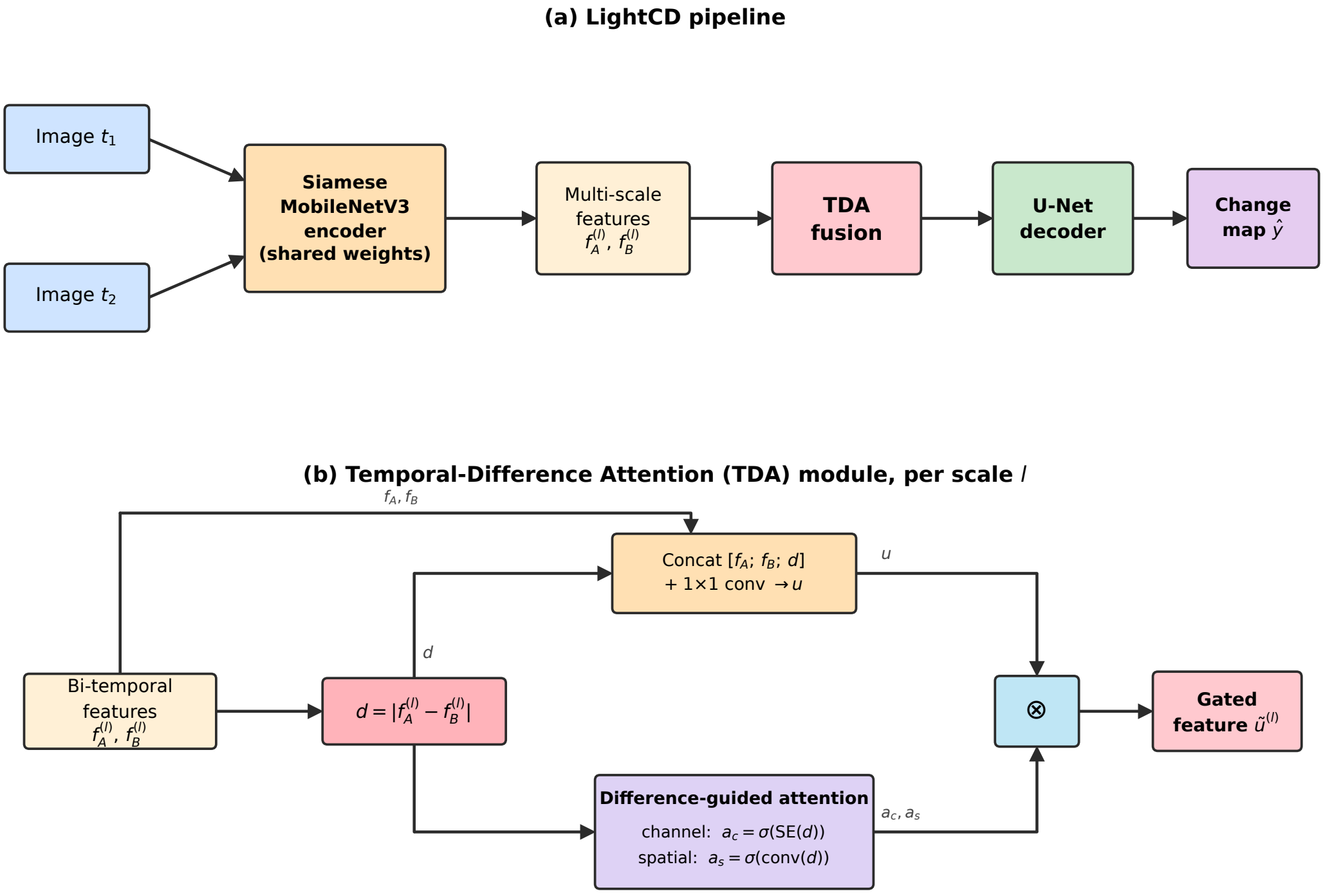
(a) LightCD pipeline
Image t_1
Image t_2
Siamese MobileNetV3 encoder (shared weights)
Multi-scale features $f_A^{(l)}$, $f_B^{(l)}$
TDA fusion
U-Net decoder
Change map $\hat{y}$
(b) Temporal-Difference Attention (TDA) module, per scale l
f_A, f_B
Concat $[f_A; f_B; d]$ + 1×1 conv $\rightarrow u$
u
d
Bi-temporal features $f_A^{(l)}$, $f_B^{(l)}$
$d = |f_A^{(l)} - f_B^{(l)}|$
$\otimes$
Gated feature $\tilde{u}^{(l)}$
Difference-guided attention
channel: $a_c = \sigma(\mathrm{SE}(d))$
spatial: $a_s = \sigma(\mathrm{conv}(d))$
a_c, a_s